

DEPARTAMENTO DE DESARROLLO HUMANO

UNIVERSIDAD YACHAY TECH

Andrés Tirado, Ph.D.

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CLASS 01: INTRODUCTION OF THE COURSE

SCIENCE COMMUNICATION



- **Professor**
- **Course Schedule**
- **Course Objectives**
- **Students´ Evaluation**
- **Course Syllabus**
- **Bibliography**

Professor



Estudios Postgrado (U. Complutense de Madrid):

- **Ph.D.: Comunicación Audiovisual, Publicidad y Relaciones Públicas.**

- **Máster en Liderazgo Democrático y Comunicación Política.**
- **Máster en Estudios Avanzados en Educación Social.**

Estudios Pregrado (U. Central del Ecuador):

- **Lic. Comunicación Social**

Otros estudios (OEA-FLACSO):

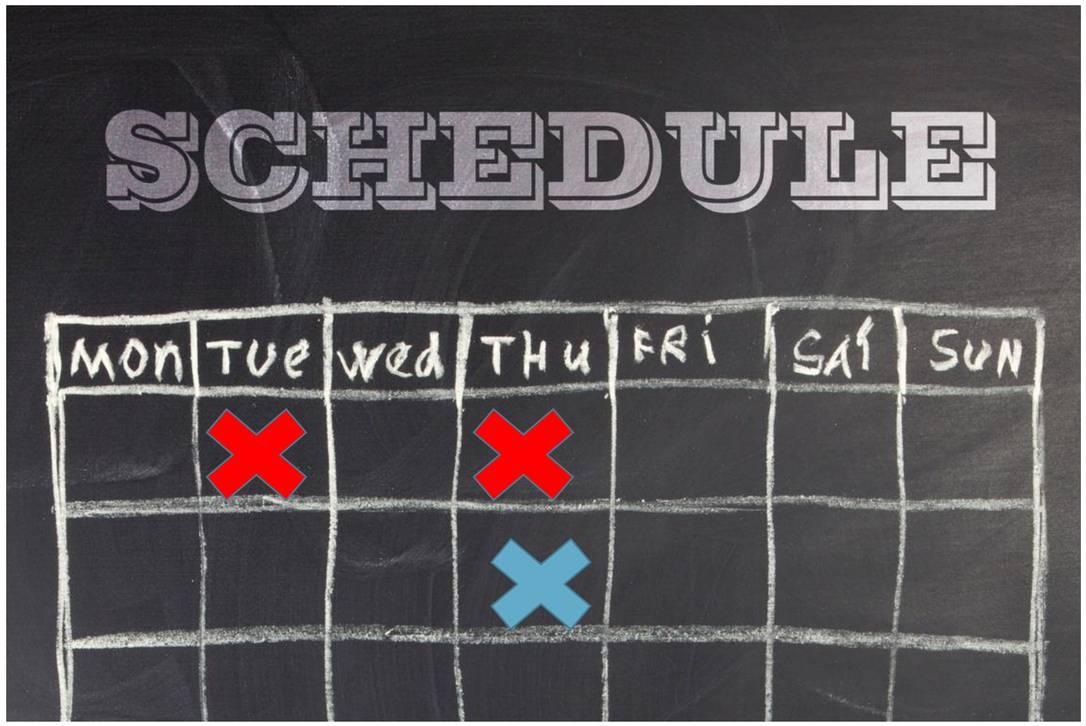
- **Diplomado en Procesos Electorales de las Américas.**

Organización de los Estados Americanos

COURSE SCHEDULE

From August 12 to December 03 of 2025

- Tuesday: 09h00 to 11:00
- Thursday: 09:00 to 11:00



TUTORING SCHEDULE:

- Thursday: 12h00 to 13:00

COURSE OBJECTIVES

The goal of this course is that you are becoming capable of effectively communicating scientific results: in both oral and written formats.

- Learn the basic structure of written and oral scientific presentations.
- Develop skills in writing and oral presentations.
- Acquire skills to explore research topics, teamwork, rational decision making and critical thinking.
- Develop key skills to write scientific articles in an affective way, that is, summarize information and systematize it effectively.
- Collect and analyze literature that addresses a specific question, problem, or area of research.
- Develop confidence to share and discuss research ideas.
- Effectively practice scientific writing and oral presentations.



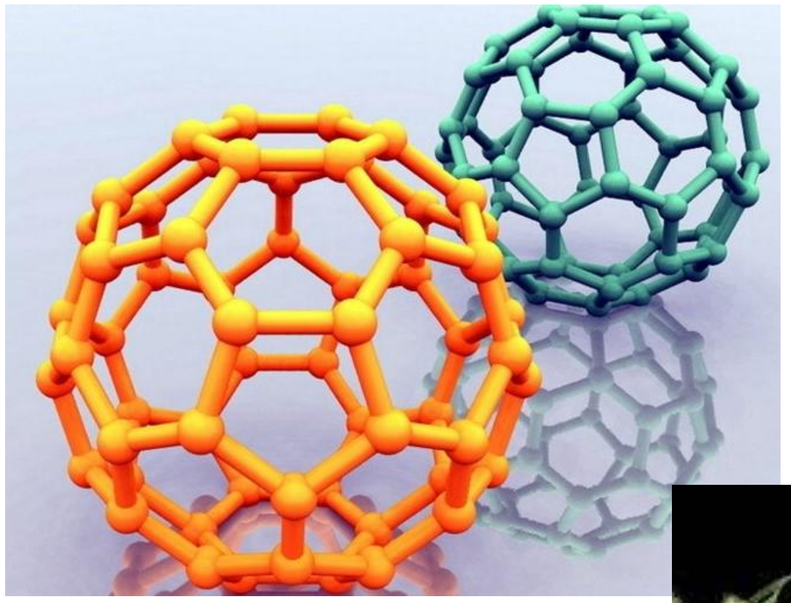
COURSE EVALUATION:

EVALUATION		
40%	**FORMATIVE**	
	20%	Formative 1: Homework, Research Project and Assistance.
	20%	Formative 2: Homework, Research Project and Assistance.
30%	**MIDTERM**	
	30%	Exam (1)
30%	**FINAL**	
	30%	Exam (1)

SYLLABUS:

	NUMBER OF WEEKS	TOPICS	ACTIVITIES
1	4 weeks	**Unit 1 – Science Communication**	• **Weekly assignment.** • **Reading.** • **Final Proyect** • **Midterm** • **Final exam**
2	7 weeks	**Unit 2 – Structure of scientific research and its applicability.**	
3	5 weeks	**Unit 3 – Written and oral skills of scientific communication**	

Final Project:



Research lines





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¡GRACIAS!

ESCANÉAME

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UNIVERSIDAD YACHAY TECH

DEPARTAMENTO DE DESARROLLO HUMANO

Communication parts and other concepts

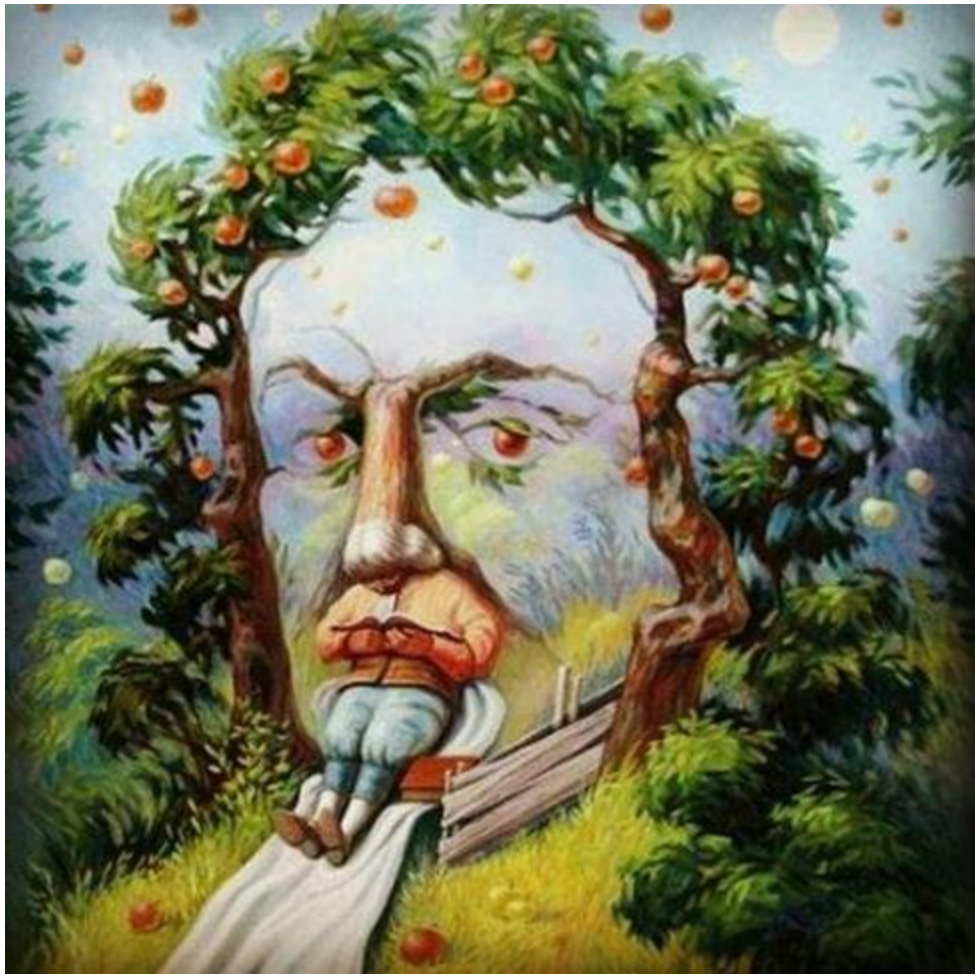
Andrés Tirado, Ph.D.

Question



What is communication?

What does this image communicate?



Communication processes



It is a process that consists of the exchange of messages between two or more people.

Activity transversal to all life processes, understood as a voluntary act.

Humans replicate this process through technology, computer systems, etc.

Communication characteristics

Communication



It is a process of information exchange, which occurs between living beings (or between cybernetic systems that imitate them). There are many different ways of communication.

Throughout the communication process, **different instances called "elements of communication" are involved**, and they must be present so that communication can take place effectively.

It generally occurs cyclically, that is, it feeds back, although it can also be unidirectional.

It can have very different general purposes, but **it is almost always about eliciting a desired response** from the interlocutor or interlocutors.

Importance of communication



There isn´t form of life that does not communicate in one way or another both with its peers and with the outside world, and therefore we are talking about a fundamental process in the known world.

T**echnology has designed mechanisms to amplify, characterize or modulate communication**, overcoming great spatial and even temporal distances, so that it is an activity of central importance in human existence.

Types of communication

Depending on the basic elements involved in it, we can classify any form of communication based on the following criteria:

Verbal and non-verbal communication. Depending on whether we use a linguistic code (that is, a language) or not, we can distinguish between:

- **Verbal communication:** Use articulate linguistic words and signs.
- **Non-verbal communication:** Use other types of signs, such as called "body language“, kinesic and proxemic.



Types of communication

The human body is a communication tool. Through dance or play, it makes people develop capacities and aptitudes.

It is a way of creating new concepts to know the different forms of communication.

Developing this knowledge always results in an increase in efficiency but also in the perception that the individual has of himself.

It is one of the basic forms for non-verbal communication. Sometimes the gestures or movements of the hands or arms can be a guide to your subconscious thoughts or emotions.



Games



- It has the ability to provide different perspectives of reality.

- It allows the relationship of people with the world.

- Tests skills and abilities.

- Has the ability to create, pursue a goal.

- People show their true essence (children's imagination is infinite).

- Character repetition and renewal



Kinesic



Palo Alto School has deepened the study of non-verbal communication.

Birdwhistell (1979) carried out studies on body micro-movements, starting from interaction and cybernetics. This cultural study is what is known as kinesics.

Kine and kinemas (50 to 60). The culture that gives it that meaning.

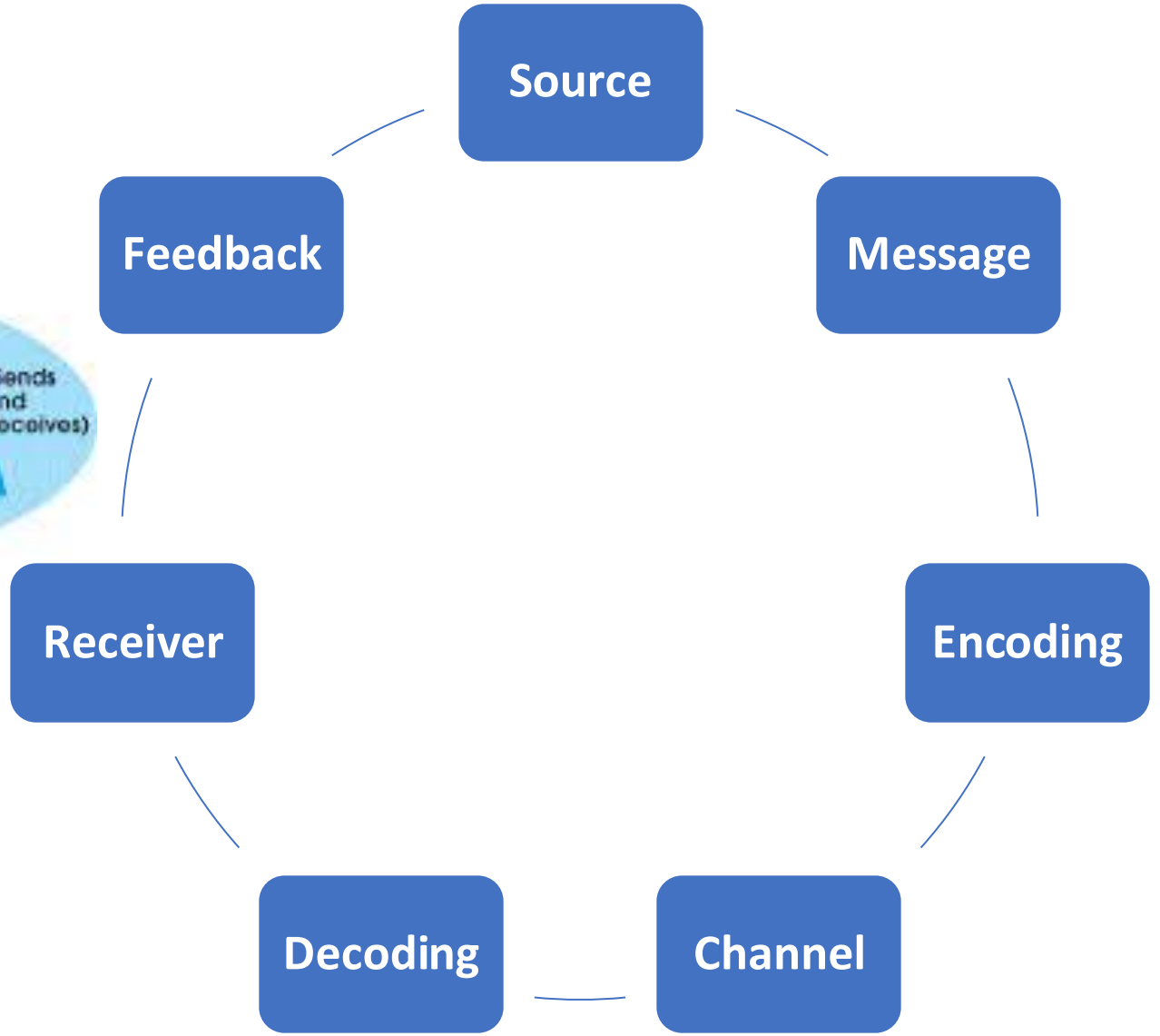
Proxemic



- The Palo Alto School also developed the study of distance and body space as a way to regulate social interactions..
- Edward Hall (1987) Intimate - social - public distance.
- Erving Goffman (1971, 1979, 199) Theory of the "personal front" natural or intentional performance.

Communication elements





Source

The one who initiates the communicative act, creating, coding and transmitting a message.

It is the one that emits a message destined to transmit or inform something in particular. It is an indispensable part of the communication process.

They must have the ability to craft a message and convey it in such a way that it is available.

It must be easily understood by the recipient. Your role is essential because if you do not carry out this management effectively, the process will probably not be carried out in a positive way.



Source

Credibility: the source must have credit before the receiver, for which personality, position within the entity and experience are important factors.

Attractive: The more “attractive” the sender's image is to the perceiver, the more effective the message will be for the latter.

Persuasion: The sender must also be persuasive enough (emotionally or rationally), so that the receiver makes the content of the message his own.

Message



The information that is transmitted, whatever the type.

It is the object of communication. It is defined as the information or verbal statement that the sender sends to the receiver through a communication channel.

It is the raison to be of the communicative process and is, at the same time, what is communicated.

You need a system of signs (letters, characters) and a code or language (Spanish, English, Portuguese).

It is transmitted orally, in writing or through images.

The form and the substance can vary, either to enhance the content of the message, or to attenuate it, modify it or rethink it, but this can also be done by the context.

Message: types

Clarity: messages must be clear, understandable and unambiguous.

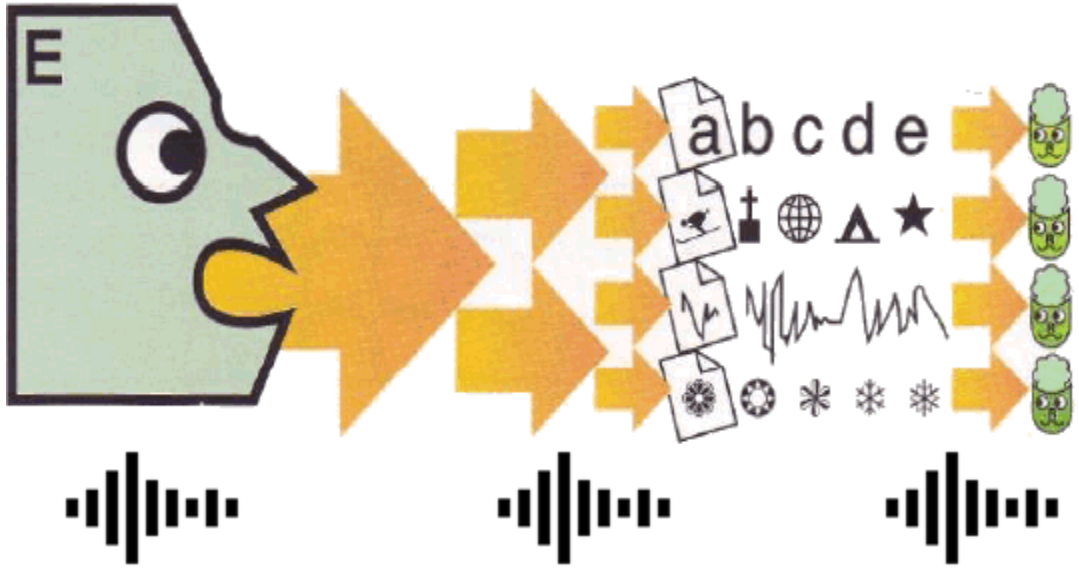
Accuracy: the information transmitted has to be complete and accurate, so that you do not believe the recipients are suspicious and suspicious.

Objectivity and veracity: the information must be true, authentic, impartial and essentially objective.

Timely: the message must be delivered when it is useful and necessary, and not before or after.

Interesting: the message must create a positive reaction in the receiver and that is only possible if it is interesting.

Code: encoding and decoding



It is a form of encryption, that is, a language or system of meaning that serves to translate the message.

In the case of verbal communication, this is very evident, since we require a shared language to transmit a message to the receiver.

A set of elements that are combined following certain rules and that are semantically interpretable.

Codes can be other types of sense systems, much more primitive, such as the one that associates grunting with risk of bite, usually on the basis of past experience.

Code: encoding and decoding

Encoding

It is a process by which the sender converts his ideas into physical signs that can be received by the receiver.

Ideas can be encoded in images, gestures, sounds, etc.

Decoding

The receiver transforms the symbolic code sent by the sender into ideas.

It is the interpretation of symbols.

Channel



This is the way by which data, messages or information sent by a sender to a receiver are transmitted.

This is the physical medium used to transmit the message. In other words, the physical way in which we carry out the communicative act: the sound waves that carry the spoken word, the ink letters on a paper, or the chemical impulses between one cell and another.

Channel: types

Personal channels: these allow more direct communication between sender and receiver and greater interaction during the process.

Impersonal channels: these communication channels have the peculiarity that they are directed to a large number of recipients. The messages are usually not personalized.

One-way channel: the media, the boom in new technologies and the appearance of the Internet are channels through which messages are broadcast to a large audience.

Bidirectional channel: when communication is carried out through a channel that allows constant and direct interaction between sender and receiver, it is called a bidirectional channel.

Channel: types

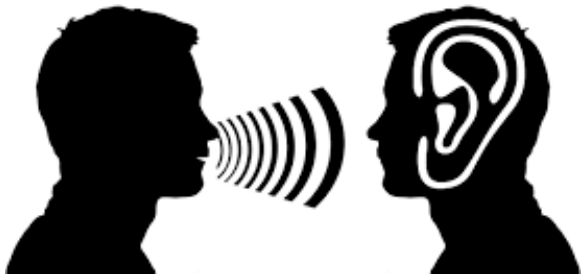
Thought or projected message: the one that the sender tries to transmit.

Transmitted message: the one actually transmitted and that is on the communication channel.

Message received: the one decoded and understood by the receiver.

In effective communication, these three messages should be identical, but in practice they rarely are. The sender can have very clear in his mind the message he wants to transmit, however, at the moment of encoding it, he can use ambiguous or incomprehensible signs for the receiver.

Receiver



The one who gets the message and is able to decode and understand it.

It is the person to whom the communication is directed.

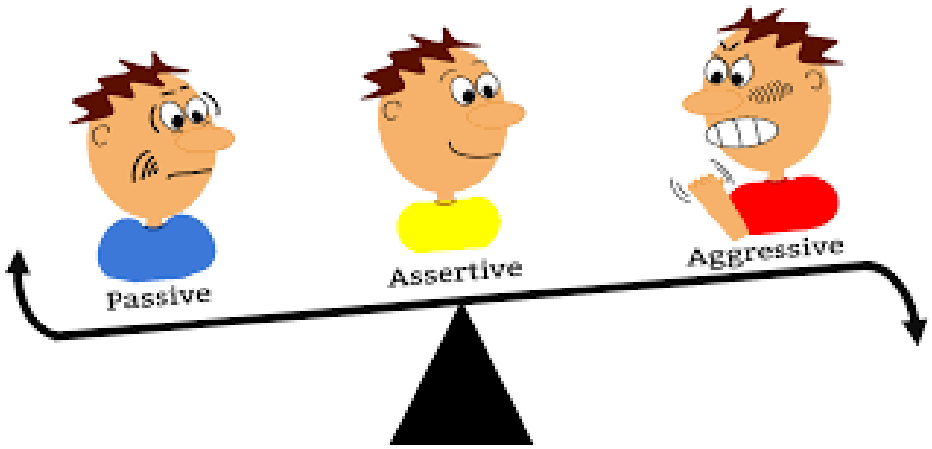
It performs a process inverse to that of the source, since it deciphers and interprets the signs used by the source. It decodes the message.

Also here we may be referring to human beings, living beings or devices with the ability to decode messages and even interpret and respond.

Assertive communication

Allows communication between individuals to be effective and the message to be clear.

- **Facilitates healthy relationships** by expressing thoughts and feelings clearly, avoiding misunderstandings and unnecessary conflicts.
- **Prevents and resolves conflicts** by allowing the appropriate expression of opinions and needs, promoting a harmonious and collaborative environment.
- **Promotes mutual respect** by considering our rights and those of others, creating an atmosphere of understanding and respect.
- **Improves self-esteem** by empowering us, reinforcing confidence by expressing ideas and feelings in a clear and respectful manner.



Other elements: Context and Feedback

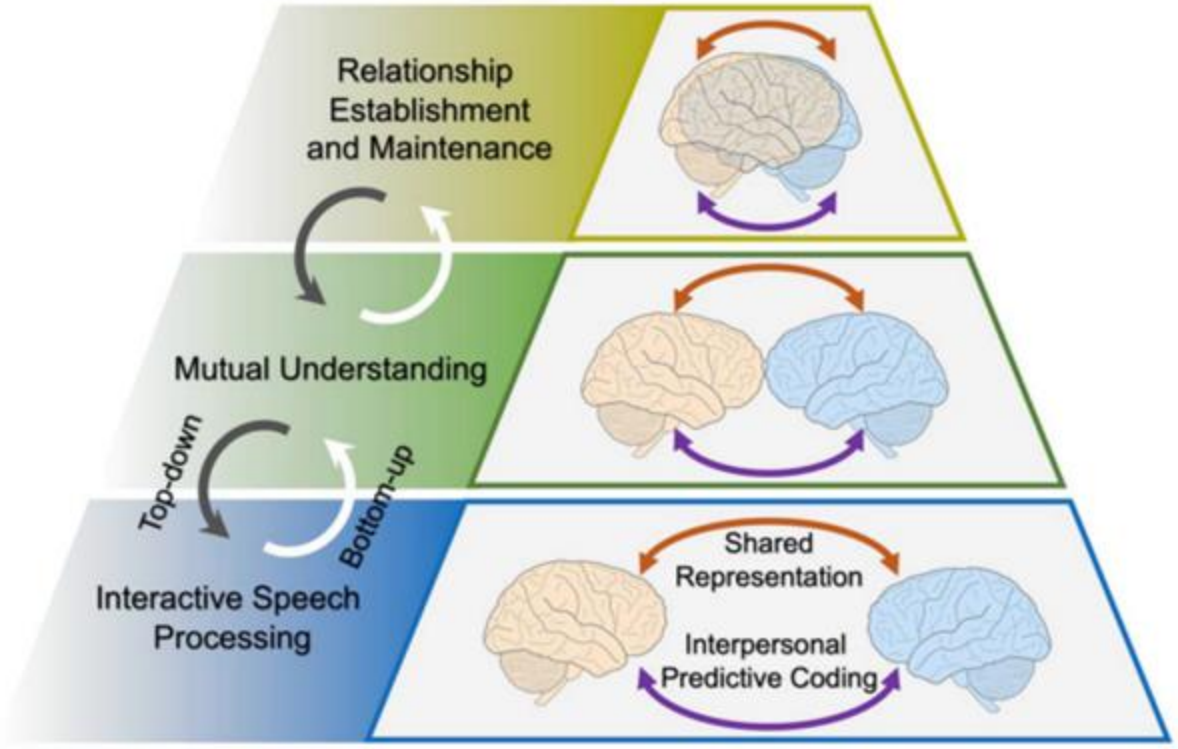
Context: it is the extralinguistic environment where the communicative act itself takes place.

- Situational contexts: these are political, historical moments, etc.
- Thematic contexts: called referents, which are constituted by the topic, subject or subject matter.



Feedback carries wi throughout the communication process. In such a way that, the sender and the receiver are successively changing roles. This makes the role of each one unclear at times.

Interpersonal verbal communication model.



1. **Interactive Speech Processing**

Involves the fundamental processing of spoken language, serving as the foundation for verbal communication.

- Examines how the human brain adapts to and evolves in response to speech.

1. **Mutual Understanding**

Concentrates on the shared understanding between communicating partners.

- Creation of a common mental representation of communicated information.
- Predicting partner's intentions, beliefs, and actions based on shared representation.

1. **Relationship Establishment and Maintenance.**

Emphasizes the social dimension of verbal communication.

- The quality of communication and relationship closeness impact the effectiveness of verbal communication at this level.
- Examples: Emotional bonding, trust, and cooperation.

Video



García (28, junio, 2021) [TEDx Talks]. Cómo la comunicación no verbal nos ayudará a ser mejores persona [Archivo de video]. https://www.youtube.com/watch?v=CY5wnAZw1-w

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¡Thank you!

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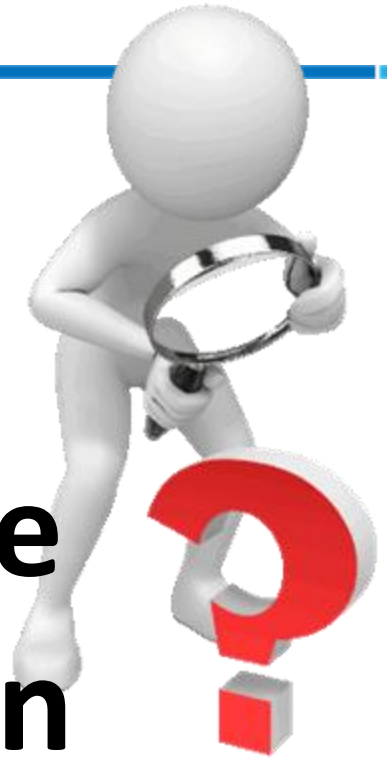
UNIVERSIDAD YACHAY TECH

Science Communication

Andrés Tirado, Ph.D.

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Question



What is science communication

Science Communication



It is a set of activities associated with the production and use of information, from the moment the scientist conceives an idea until the information on the research results is published.

Science Communication

The main objective of scientific communication is the integral management of knowledge including:

- Knowledge Registration
- Rigorous Evaluation
- Dissemination
- Knowledge Accumulation

Focus on Facts and Human Perceptions

Science Communication: general context

The Declaration on Science and the Use of Scientific Knowledge adopted in the UNESCO sponsored World Conference on Science in 1999, established the following:



1. Scientific knowledge should be shared.

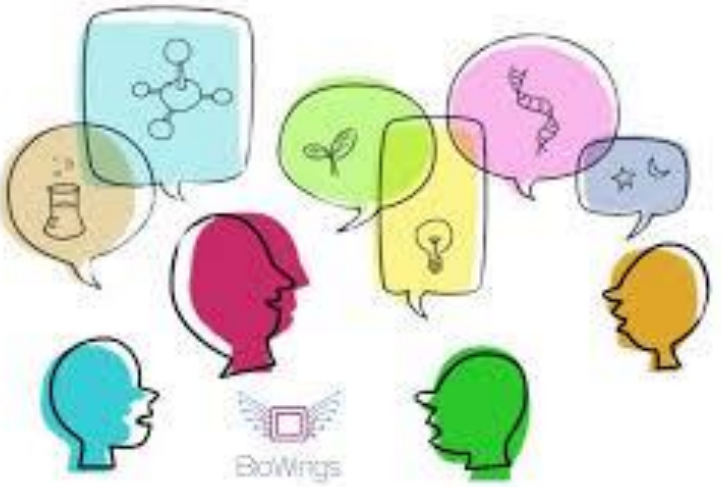
2. There is need for real cooperation between governments, civil society, the business sector and scientists.

3. Scientists should be governed by ethical standards.

4. Professionals who circulate scientific information see communicating with the public as a vital responsibility.

Dissemination and Disclosure

The dissemination of scientific knowledge allows research results to be known, discussed and shared with the academic community.





Disclosure refers to the act of making information known or public, especially information that was previously private or unknown.

Scientific communication: dissemination

Features

Provide the scientist with the best answers to specific questions.

Contribute so that the scientist knows about the new developments in his field of action.

Encourage the scientist to take new knowledge, in addition to that of his area of interest.

Disseminate the main trends in emerging areas, providing scientists with an idea of the relevance of their work.

Scientific communication: dissemination

Features

Test the reliability of the new knowledge, before the possibility of testimonies and verifications.

Redirect or expand the role and interest of scientists.

Provide feedback for the improvement of the researcher's production.

Science communication: disclosure

Features

The production of knowledge occurs from previous results.

Researchers consume and communicate information from the beginning to the end of their studies.

Science depends on the wide dissemination of what is produced.

Science depends on communication processes.

Science communication: disclosure

Feature	
	Promote curiosity towards a subject or field of knowledge.
	Offer the information necessary for society to acquire an opinion on issues related to advances in science and their repercussions.
	Help understand the social, political or cultural transformations of new events and facilitate each person to have critical thinking.

What is scientific disclosure for?



Reduces the educational gap in the different population sectors

It works as an educational element that facilitates the population's access to academic, technical or specialized information.

People can form an opinion on a controversial or difficult issue, and be able to act on it.

La Presentación de tu Vida (29, septiembre, 2019). Cómo hacer DIVULGACIÓN CIENTÍFICA [Archivo de video]. Recuperado de: https://www.youtube.com/watch?v=UWDsWwG0BAI

Dissemination versus scientific disclosure

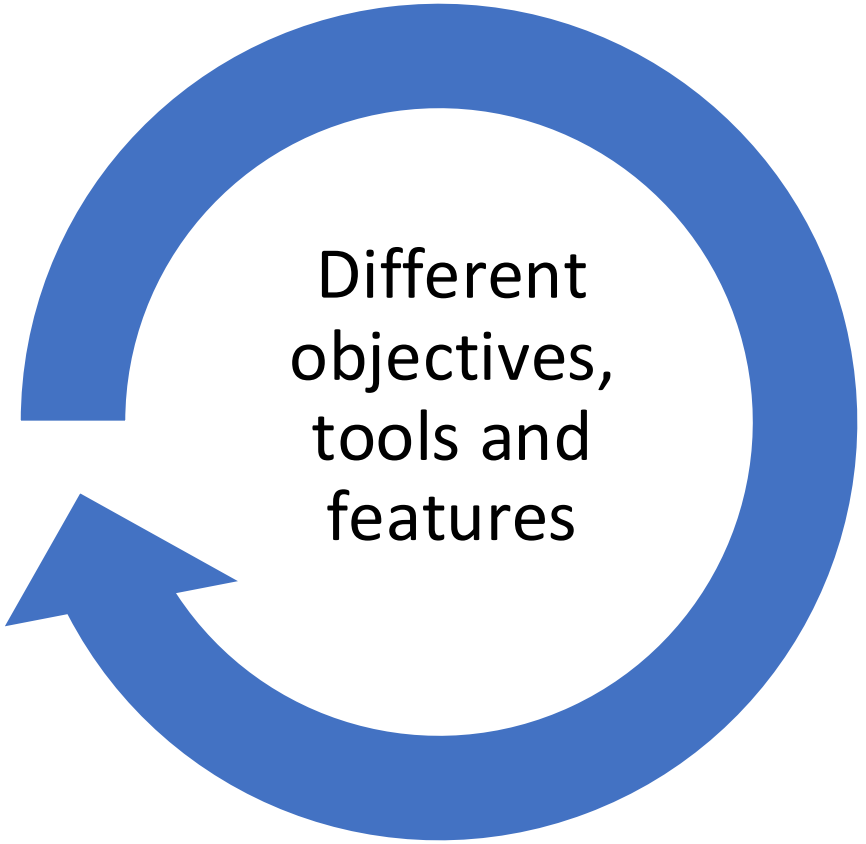
Dissemination	Disclosure
Communication between "peers", that is, between specialists.	Communication between "non-peers", partners do not share the same knowledge.
It establishes a symmetrical relationship between the subject who communicates and the individual who interprets.	Interdisciplinary work that communicates knowledge to various audiences.
It proposes a delimited and specific socio-professional or technical framework.	General audience.

Who should say science?

There is a consensus on the need to **bring scientific knowledge to society** and get educated and informed citizens, who support science and its financing and have sufficient criteria and opinion on issues that affect everyone.



- **Scientists**
  Journalists
  Disclosers
  Institutions / companies



Scientists

Credibility | Knowledge | Authority



They assume an essential role in the dissemination of science, since they are the top experts in their specialty.

- They have found outreach to be an interesting and fun activity, they consider it part of their job and do it on a regular basis.
- They use new technologies, they communicate quickly, directly and on a daily basis.
- Dissemination channels: blogs, social networks, articles, conferences and books.

Journalists

- Diffusion channels: traditional media (TV, press and radio), digital media, blogs, podcast and social networks.
- Professionals with the ability to tell attractive stories that demonstrate to citizens the consequences of scientific knowledge in their lives.
- It is responsible for translating knowledge so that people can learn clearly.



Historically they are the actors in the dissemination of science.

Scientific disseminator

Scientific dissemination can be carried out by **non-scientific disseminators and scientific disseminators.**

In either case, publishers must maintain these principles in their disclosure product:

- Be faithful to the scientific content.
- Be aimed at a **non-specialized audience.**
- Create a new message with **non-technical and contextualized language** to make it accessible.
- Express oneself under the agreement of a **voluntary public.**

Scientific disseminator



B A L A N C E

The essential tension of science popularization is based on these principles:

A) A product close to science requires technical language, therefore it needs more context.

B) A product with pleasant language has a probability of errors.

Companies / institutions

Institutions related to science.



Companies in the technological or innovative field.



- They are important actors in the popularization of science.
- Its mission is to provide any interested party with immediate and easy access to information and documentation in their area.
- They use the web pages, open houses and other activities.

What does the media work for?



Human beings need the media to understand the world around them.

They are social animals and are increasingly interested in knowing what happens beyond their professional and family life.

They seek to know what happens in the political, economic, cultural, sports, etc.

Out of curiosity, but also because these areas directly or indirectly influence our lives.

Communication theories: Agenda setting

- **McCombs et al. (2014)**

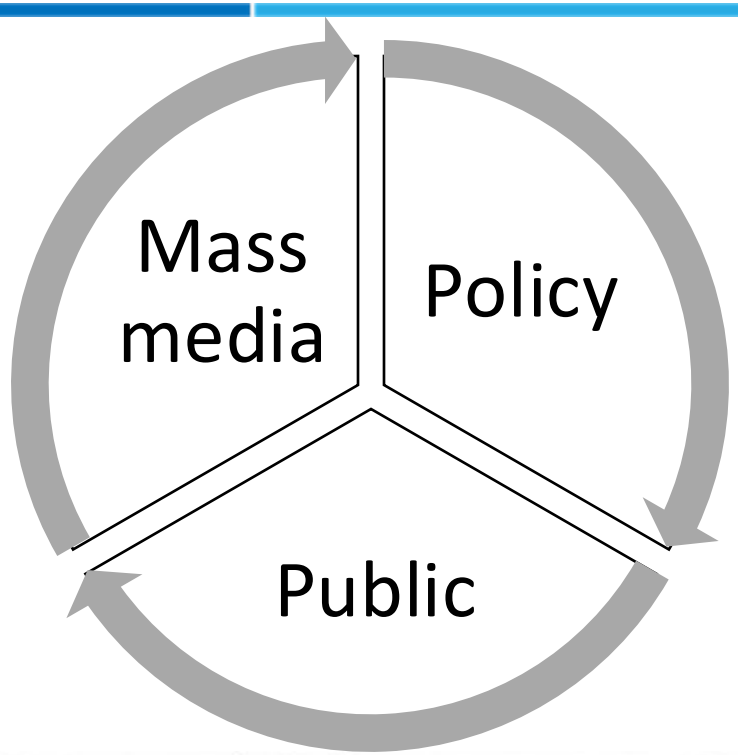
The works carried out on the Theory of the agenda-setting from different aspects reach the same outlet, in which the media have the ability to create reality.

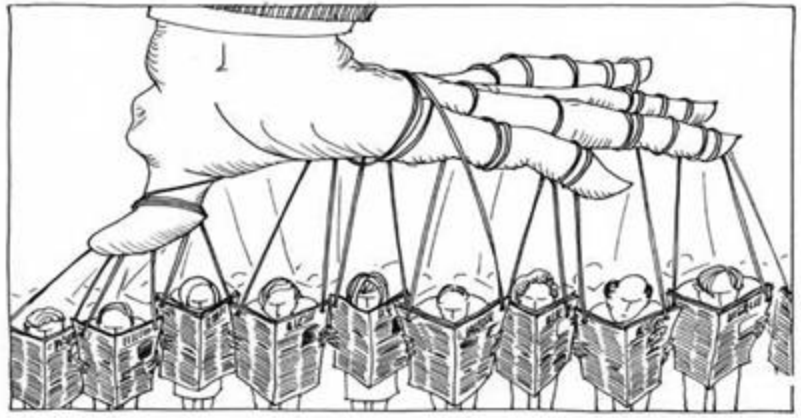
- **Santillán (2018)**

They claim that the formation of public opinion is a consequence of the role of the media.

- **Tirado-Espín, A., et al. (2020).**

The determination of a set of news makes citizens perceive that reality as the most important of its day.





Communication theories: Framing

- **McCombs y Shaw (1972)**

The news not only marks the relevance or importance of the social issues proposed by the agenda. That is why this theory is known as the second level of agenda setting theory.

- **Entman (1993)**

People's way of thinking about certain issues is manipulated, depending on how you put them. This way of presenting information is known as a news frame or frame.

- **Aday (2006) y Keum et al. (2005)**

These frames have effects on the cognitive responses of the subjects as well as on people's attitudes and beliefs about the issues discussed.



Media features

Report

The events that occur are told.

NEW

- Investigation process.
- Contrasting of sources.
- Deontological codes.
- Audiovisual and written media.



Media features

Explain

What happens in the world is necessary to explain the context in which the events occur.

REPORTAGE



Vision as wide and varied as possible of the subject in question.

The space / time available is usually much greater.

The journalist must refrain in these genres from including their own opinions directly.

Funciones de los medios de comunicación

Opine

People need to be told what is happening in the world so that they can build an opinion of that reality.

The media make the task easier by including different types of text or genres.

EDITORIAL

ARTICLE

COLUMN



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06 Alzate Zuluaga, M. L., & Romo Morales, G. (2017). La agenda pública en sus teorías y aproximaciones metodológicas. Una clasificación alternativa. *Revista Enfoques: Ciencia Política y Administración Pública*, *XV*(26), 13–35.

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08 Entman, R. M. (1993). Framing: Toward Clarification of a Fractured Paradigm. *Journal of Communication*, *43*(4), 51–58. https://doi.org/10.1111/j.1460-2466.1993.tb01304.

09 Santillán Buelna, J. R. (2018). Imágenes de México en la 'La Vanguardia' y 'El Periódico de Catalunya' (2013-2016). *Index. Comunicación: Revista Científica En El Ámbito de La Comunicación Aplicada*, *8*(1), 101–121.

¡Thank you!

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DEPARTAMENTO DE DESARROLLO HUMANO

UNIVERSIDAD YACHAY TECH

Scientific dissemination

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Scientific communication channels

Formal channels

They are communications of a public and lasting nature.

The material is available to a wide audience.

Scientific journals, books, reports and proceedings.

Informal channels

Ephemeral forms of communication.

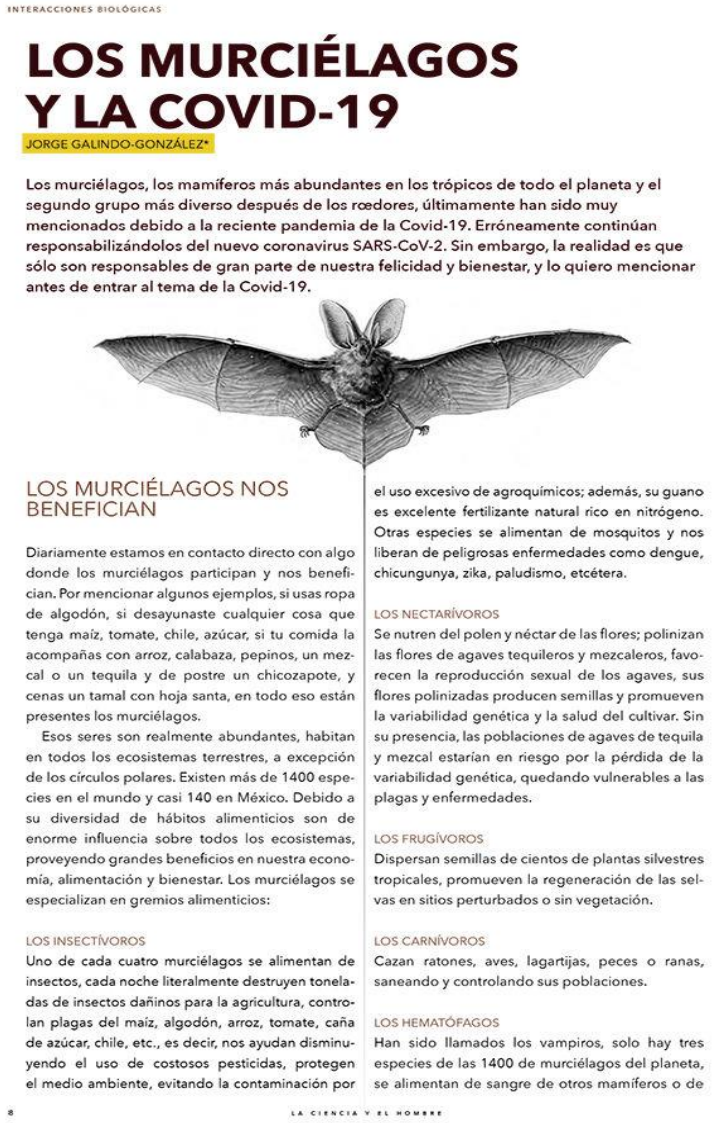
They are presented to a limited audience and for a short time.

Scientific talks and meetings.

Scientific communication channels

Formal channels	Informal channels
Potentially large audience.	Restricted public with limited access.
Information can be permanently saved and retrieved.	Information is not generally saved or retrieved.
The information is relatively old.	The information is recent.
Selection of the channel and content is the initiative of the user.	Selection of the channel and content is the initiative of the user.
Little feedback for the author.	A lot of feedback for the author.

Scientific dissemination articles



INTERACCIONES BIOLÓGICAS

LOS MURCIÉLAGOS Y LA COVID-19

JORGE GALINDO-GONZÁLEZ*

Los murciélagos, los mamíferos más abundantes en los trópicos de todo el planeta y el segundo grupo más diverso después de los roedores, últimamente han sido muy mencionados debido a la reciente pandemia de la Covid-19. Erróneamente continúan responsabilizándolos del nuevo coronavirus SARS-CoV-2. Sin embargo, la realidad es que sólo son responsables de gran parte de nuestra felicidad y bienestar, y lo quiero mencionar antes de entrar al tema de la Covid-19.

LOS MURCIÉLAGOS NOS BENEFICIAN

Diariamente estamos en contacto directo con algo donde los murciélagos participan y nos benefician. Por mencionar algunos ejemplos, si usas ropa de algodón, si desayunaste cualquier cosa que tenga maíz, tomate, chile, azúcar, si tu comida la acompañas con arroz, calabaza, pepinos, un mezcal o un tequila y de postre un chicozapote, y cenas un tamal con hoja santa, en todo eso están presentes los murciélagos.

Esos seres son realmente abundantes, habitan en todos los ecosistemas terrestres, a excepción de los círculos polares. Existen más de 1400 especies en el mundo y casi 140 en México. Debido a su diversidad de hábitos alimenticios son de enorme influencia sobre todos los ecosistemas, proveyendo grandes beneficios en nuestra economía, alimentación y bienestar. Los murciélagos se especializan en gremios alimenticios:

LOS INSECTÍVOROS

Uno de cada cuatro murciélagos se alimentan de insectos, cada noche literalmente destruyen toneladas de insectos dañinos para la agricultura, controlan plagas del maíz, algodón, arroz, tomate, caña de azúcar, chile, etc., es decir, nos ayudan disminuyendo el uso de costosos pesticidas, protegen el medio ambiente, evitando la contaminación por el uso excesivo de agroquímicos; además, su guano es excelente fertilizante natural rico en nitrógeno. Otras especies se alimentan de mosquitos y nos liberan de peligrosas enfermedades como dengue, chicungunya, zika, paludismo, etcétera.

LOS NECTARÍVOROS

Se nutren del polen y néctar de las flores; polinizan las flores de agaves tequileros y mezcaleros, favorecen la reproducción sexual de los agaves, sus flores polinizadas producen semillas y promueven la variabilidad genética y la salud del cultivar. Sin su presencia, las poblaciones de agaves de tequila y mezcal estarían en riesgo por la pérdida de la variabilidad genética, quedando vulnerables a las plagas y enfermedades.

LOS FRUGÍVOROS

Dispersan semillas de cientos de plantas silvestres tropicales, promueven la regeneración de las selvas en sitios perturbados o sin vegetación.

LOS CARNÍVOROS

Cazan ratones, aves, lagartijas, peces o ranas, saneando y controlando sus poblaciones.

LOS HEMATÓFAGOS

Han sido llamados los vampiros, solo hay tres especies de las 1400 de murciélagos del planeta, se alimentan de sangre de otros mamíferos o de

8 LA CIENCIA Y EL HOMBRE

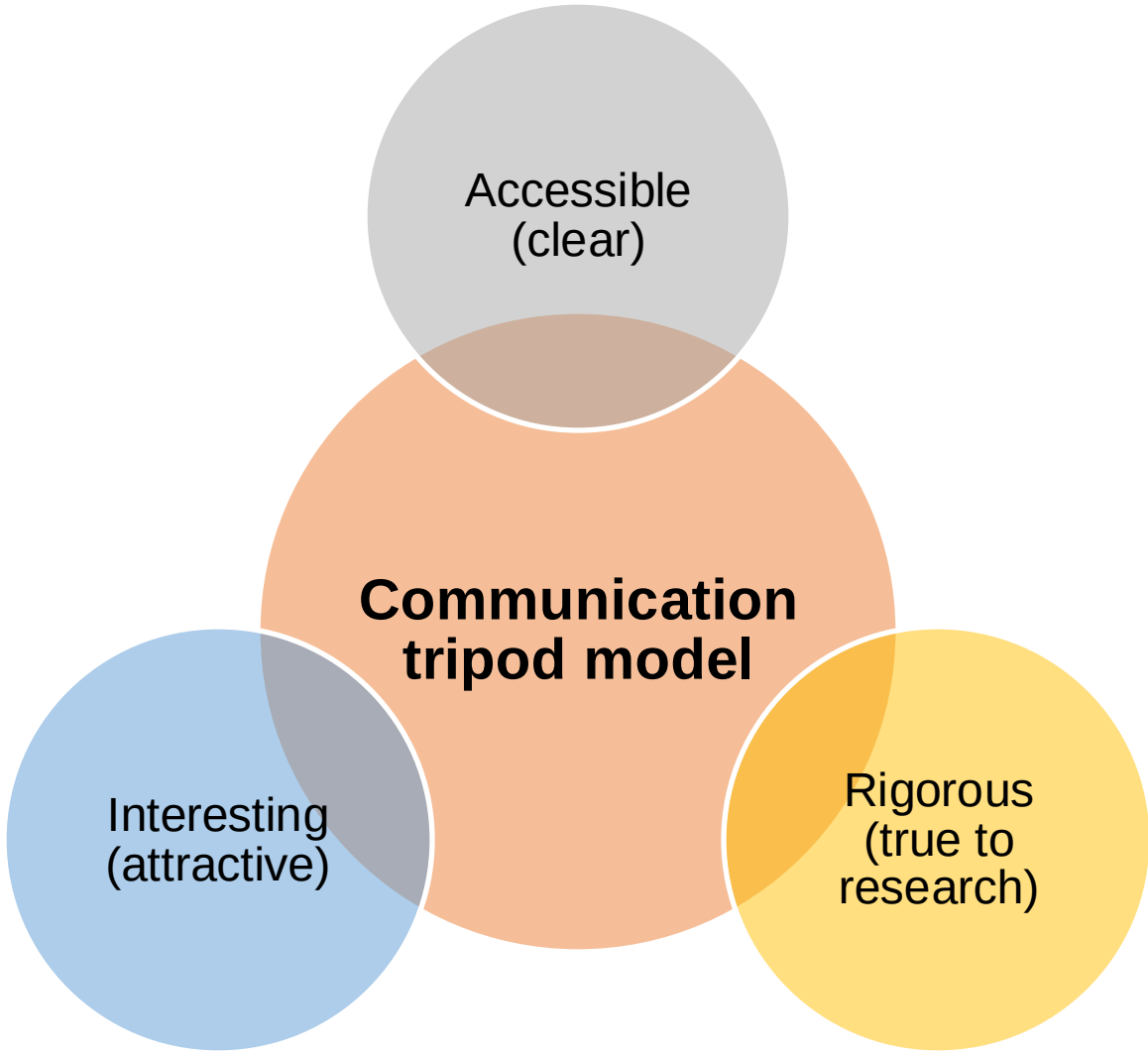
They are informative texts that are based on research projects, scientific articles (papers) or arguments that arise from science.

Its main purpose is to disseminate the research in an accessible way so that it is understandable by the general public.

Scientific dissemination articles

Topic and style

- **The topic** to be discussed must be relevant, attractive and the product of scientific research.
- **The style** must have the structure of an explanatory sequence (spread the information, explain it, interpret it and clarify the ideas).



Scientific dissemination articles

Language

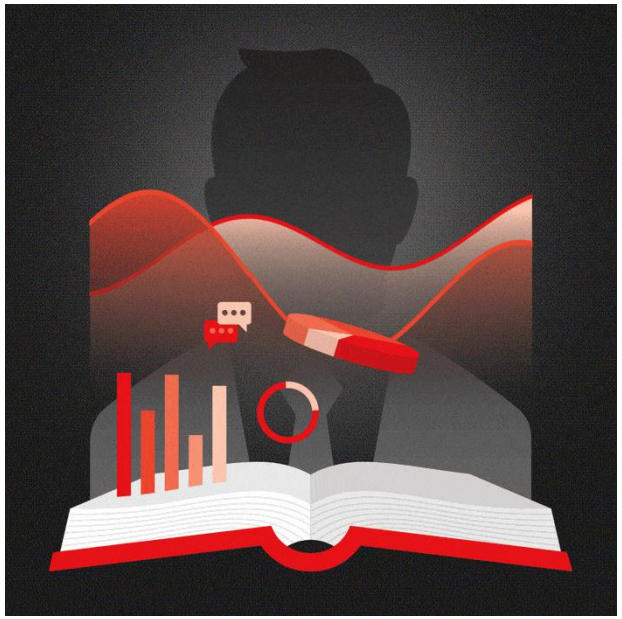
It must be agile, clear and easily understood by the non-specialized reader, avoiding the technical language of your specialty.



Storytelling

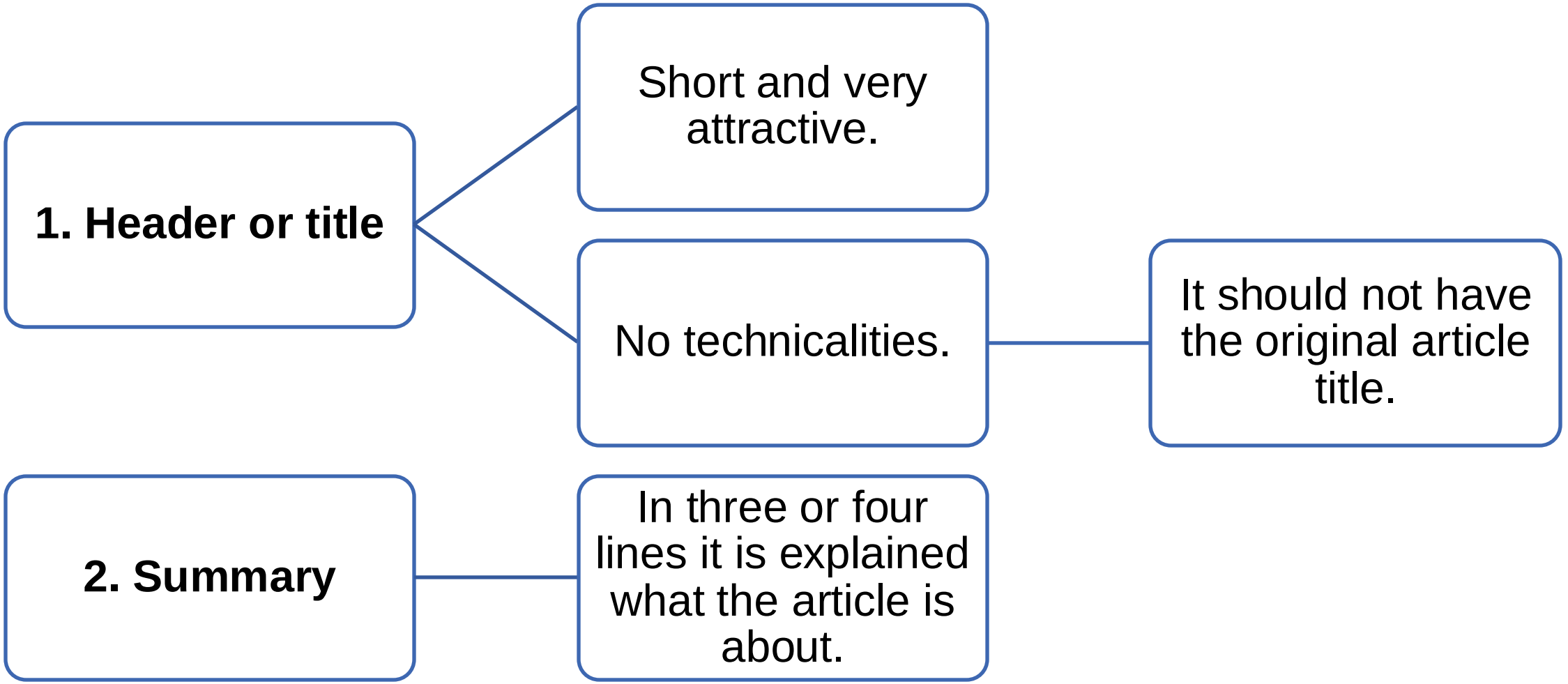
- Telling stories helps to understand the world. The narrative in science is the same.
- The dissemination article tells a story around a research topic.
- It is not about changing the facts, but about changing the way they are presented.

Parts of a dissemination article

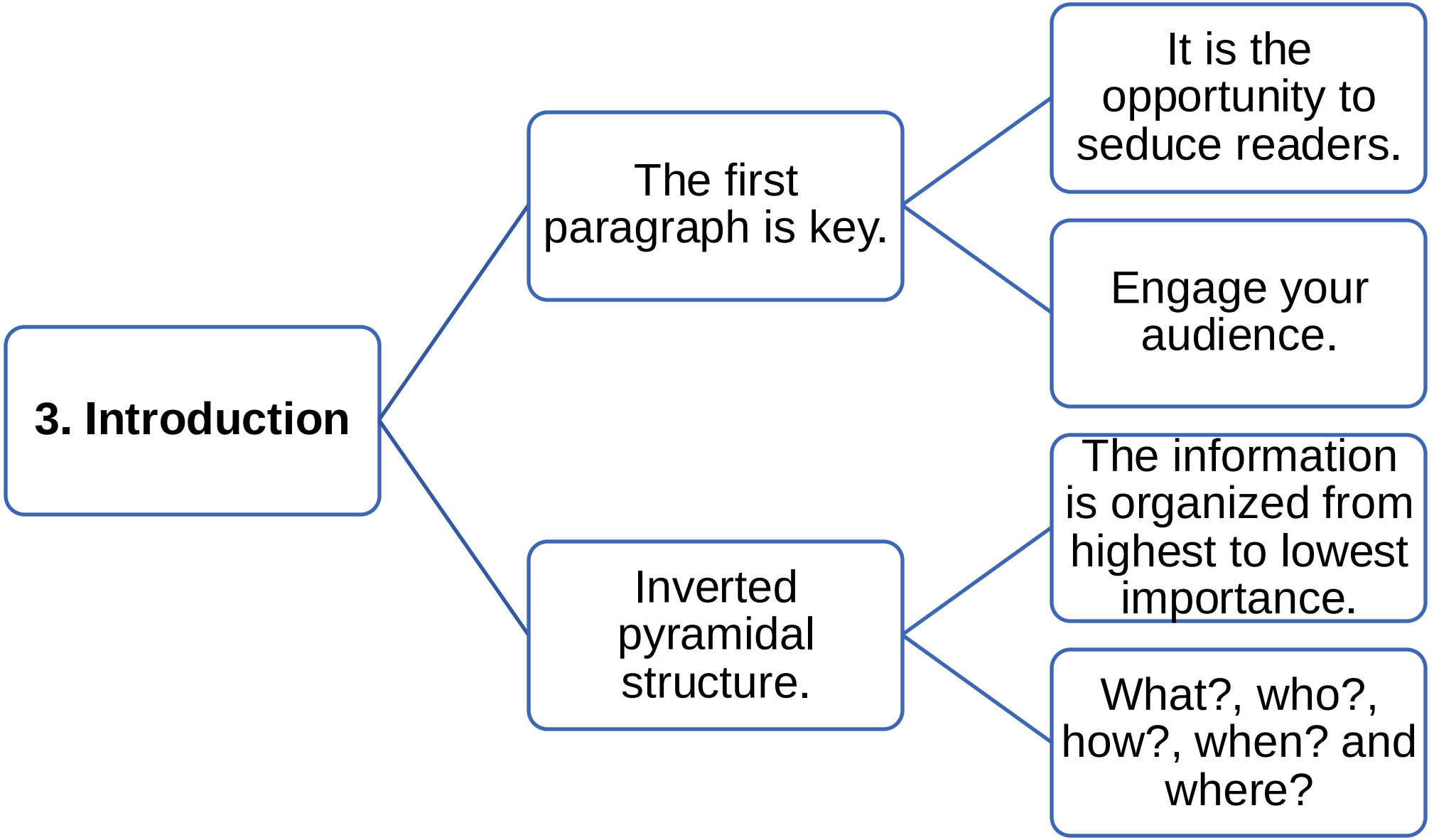


1. Header or title.
2. Summary.
3. Introduction.
4. Development.
5. Conclusions
6. Author's profile

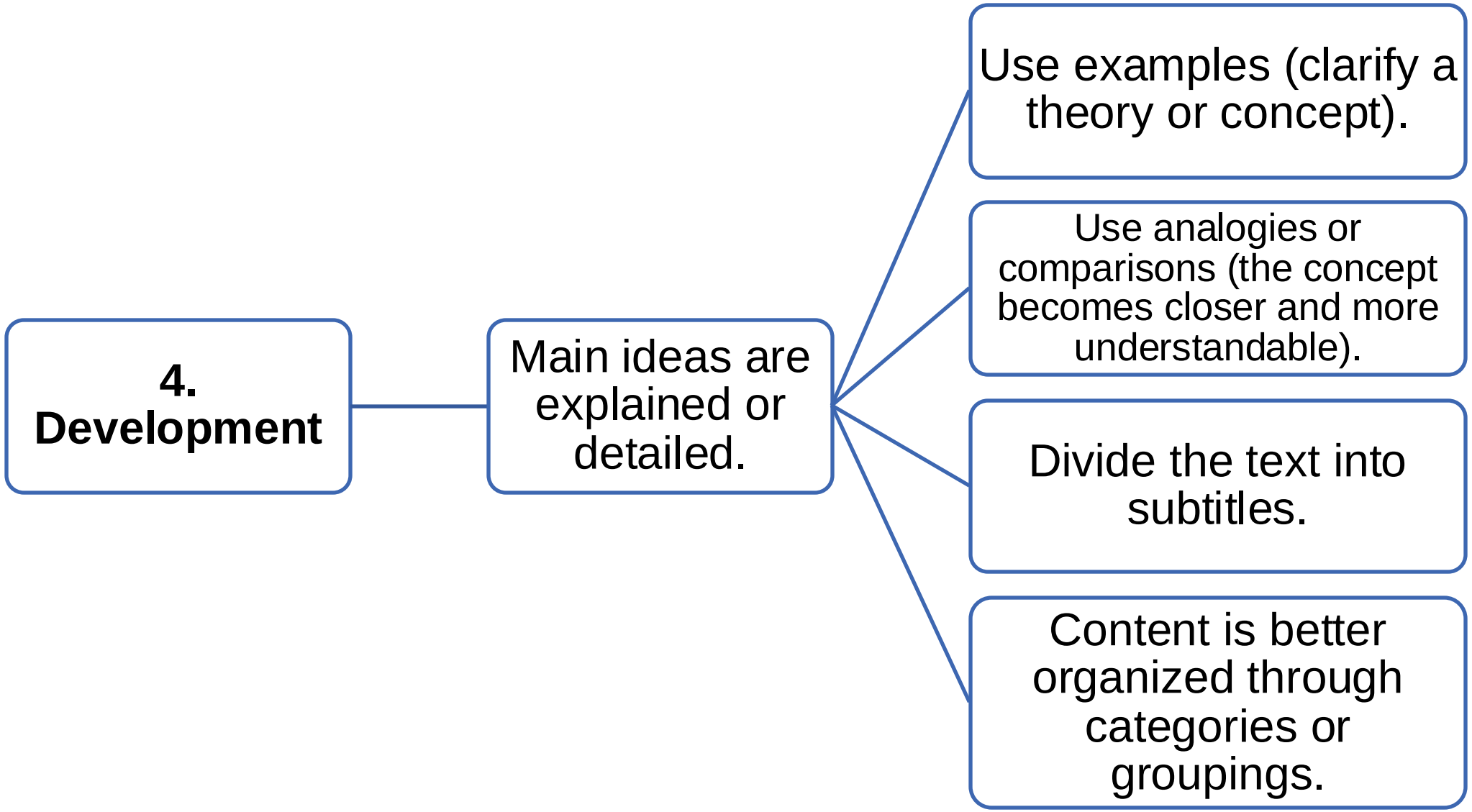
Parts of a dissemination article



Parts of a dissemination article



Parts of a dissemination article



Parts of a dissemination article

5. Conclusions

Make an evaluation on the subject.

- It expresses the possible benefits of the research.
- The closing of the writing is developed.

6. Author's profile

Identification data

- Full name.
- Email.
- Short professional review.

Scientific dissemination articles

Recommendations	Write short paragraphs.
	Use the simple structure of the statement (subject + verb + complement).
	Use synonyms, metaphors, analogies and examples to better explain a complex idea.
	Avoid excessive technicalities. If you need to use a technical term, include a simple definition.
	Humanize the subject, so that the reader can relate the advances of science with their daily life.

Scientific dissemination articles

Support resources

- The following elements may be included in the text, in order to illustrate the article of disclosure:
  - Boxes with complementary information.
  - Photographs.
    Infographics.
    Stadistic charts.
    Illustrations.
    Drawings.
    Glossaries

Scientific dissemination articles

Information photo

Title

Introduction

Capital letter

Development

Photography

Conclusions

Bibliography



AÑO INTERNACIONAL DE LA ASTRONOMÍA

DÓNDE NACEN TODAS LAS ESTRELLAS

La Gran Nebulosa de Orión es tan brillante que se puede distinguir a simple vista, incluso desde lugares con gran contaminación lumínica

La nube o nebulosa se fragmenta en multitud de pequeños cascarones y cada uno formará una estrella

DIARIO CÓRDOBA | ZOCO

Imagen de la Gran Nebulosa de Orión, tomada en un momento en que su tonalidad era roja por la mayor presencia de azufre

Storytelling

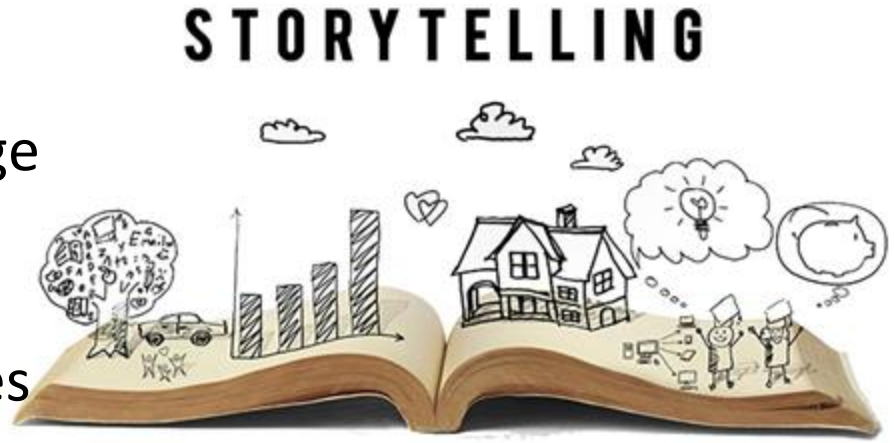
What is it?

- It is a technique that is broken down into two words: history (story) and telling (telling), through which messages are transmitted.

- It is described as a narrative of events with a final message that leaves a learning or concept.

- Experts on the subject say that knowing how to tell stories is an art.

- The goal is to achieve a larger audience, that is, more views and reproductions, but also to have an effect on the people who listen, see or read.



Storytelling

¿What is the objective?

Connect emotionally with the other through a story.

01. Captar la atención

02. Despertar emociones

03. Ganar voluntades

04. Cerrar acuerdos

05. Comprometer acciones

06. Influir y crecer



Storytelling: main elements



1. Message
2. Setting
3. Character
4. Conflict

Storytelling

1. Message

1. Story: a message is transmitted.

2. Telling: the way that message is presented to the audience.

If the message is strong, it may take effect even with a weak saying. But, if the message is poor, you will hardly be able to save your content despite using good techniques.

Texts, stories and conferences that leave the audience enthusiastic for a moment exist in piles, but contents that really impact and make you continue to remember them are few.

2. Setting

Events need a **physical space** in which to place them to continue describing the scenes.

It manages to place the protagonists and scenarios clearly. Imagination and mental translation is easier for those who hear or see the narrative in question.

Events need to happen somewhere. When the environment is well described, it enables the audience to get hooked. In such a way that, it will be from beginning to end.

Storytelling

3. Character

To put together a storytelling, the protagonist or group of people who develop the story that is told must be identified.

The character is the one who runs through the entire story and undergoes a transformation that leads to the transmission of the message.

To reach that final concept, you must overcome your conflict.

The character can be a hero or an antihero.

4. Conflict

It is the **main factor** that causes interest in the audience.

A **very simple conflict does not arouse interest**, since it does not generate identification with the public. Very easy achievements are not usually valued.

It must be elaborate and cannot have an easy solution. In this case, we would have a **romanticized story**. Although it can arouse emotions, it hardly causes identification.

Storytelling: keys to its construction

Authenticity

Lean on past events in our life to be consistent.	Create specific, well-detailed images and strong moments.

Empathy

This feeling will be the element that connects with the public.	Narrate specific examples, look for these elements in the target audience.

Don't pretend

Be authentic at all times.	Remember your body also communicates.

Storytelling: keys to its construction

Emphasis

Generate rhythms in speech, emphasize words, images and energies.	It is recommended not to be arrogant, on the contrary, find common ground with the audience.

Personalize

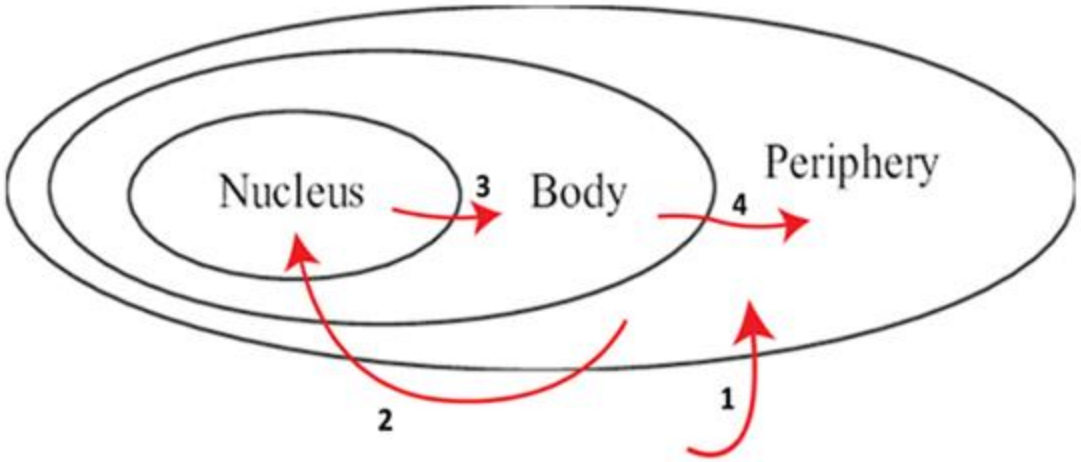
It is recommended not to be arrogant, on the contrary to find common ground with the audience.	Be humble and at the same time speak in a personal way. You must take care of the forms.

Simplify

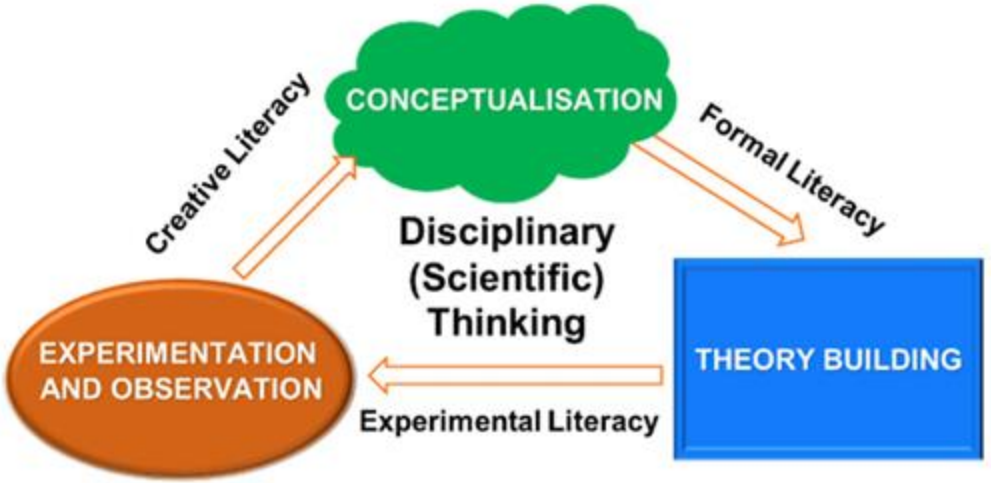
The simpler and shorter the better. Do not embellish the stories with unnecessary facts.	The stories have to have a meaning, that the beginning, middle and end are understood.

Storytelling for Culturo-Scientific Thinking

It focuses on the early development of scientific skills and modern mindset through outreach activities.



(a)



(b)

Storytelling

Conferences



Conferencias de diversos temas, dónde se pone en práctica la importancia de crear un **storytelling.**

Audiovisual production

Autoescuela Sevial-Sevilla (23, julio, 2015) La canción más cara del mundo DGT [Archivo de video]. https://www.youtube.com/watch?v=RatwBxP1V-Q

Ying Liu, R. (12, enero, 2020) Publicidad sobre el medio ambiente [Archivo de video]. https://www.youtube.com/watch?v=QcMn7JpfRVo

Written production



Referencias bibliográficas

01 Deroy Domínguez, D. (2022). *Las revistas científicas y su rol en la difusión del conocimiento científico Scientific Journals and their role in the dissemination of scientific knowledge*. https://orcid.org/0000-0002-5156-7950

02 Zhang, L., Malviya, S., Tandoc, E. C., & Ho, S. S. (2022). Exploring channels and gaps in information dissemination and acquisition among energy scientists and the public: The southeast Asian context. PLoS ONE, 17. https://doi.org/10.1371/journal.pone.0273626

03 Tirado-Espín, A., Cuesta, U., Martínez-Martínez, L., & Almeida-Galárraga, D. (2020). Agenda-setting e inmigración: análisis crítico del discurso y frecuencia en los medios: Estudio descriptivo de investigaciones en revistas científicas desde 2015 a 2020. *Revista Ibérica de Sistemas e Tecnologias de Informação*, (E35), 289-301.

04 Calua, M., Delagado, Y., & López, Ó. (2021). Comunicación Asertiva En el Contexto Educativo : Revisión Sistemática Assertive Communication in the Educational Context : Systematic Review. *Revista Boletín Redipe*, *10*(4), 315–334.

Referencias bibliográficas

05 Ashcraft, L. E., Quinn, D. A., & Brownson, R. C. (2020). Strategies for effective dissemination of research to United States policymakers: a systematic review. In Implementation Science (Vol. 15, Issue 1). BioMed Central Ltd. https://doi.org/10.1186/s13012-020-01046-3

06 Calua, M., Delagado, Y., & López, Ó. (2021). Comunicación Asertiva en el Contexto Educativo : Revisión Sistemática Assertive Communication in the Educational Context : Systematic Review. *Revista Boletín Redipe*, *10*(4), 315–334.

¡Thank you!

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DEPARTAMENTO DE DESARROLLO HUMANO

Parts of scientific research

Andrés Tirado, Ph.D.

Scientific article

What is a scientific article?

1. It is a written document that describes the original research results.

2. It has a defined structure, although some publishers propose slight changes of order and sections on the general scheme.

3. Original, written and published report that sets out and describes experimental results, new knowledge or experiences that are based on known facts.



Scientific article

Characteristics

The results must be valid and reliable (they don´t necessarily have to be experimental, they can also be theoretical).
They are original: each article communicates the results of an investigation for the first time.
They are presented in scientific journals, conferences and other modes of presentation.
It usually occupies an average of 7-8 pages of a magazine. Accompanied by graphs, tables and occasionally photographs and drawings. Although this information will depend on the rules of the magazine.

Scientific article

Characteristics	The number of authors or signatories does not usually exceed six, considering the first as the main author of the article.
	Among the authors, who is in last place is understood to be the researcher who directs the work.
	It is a formal, public, controlled and orderly document.
	Must meet key writing criteria.
	It has scientific rigor and logical character.

Scientific article

Characteristics

It must be clear and precise. The use of scientific language and vocabulary is necessary.

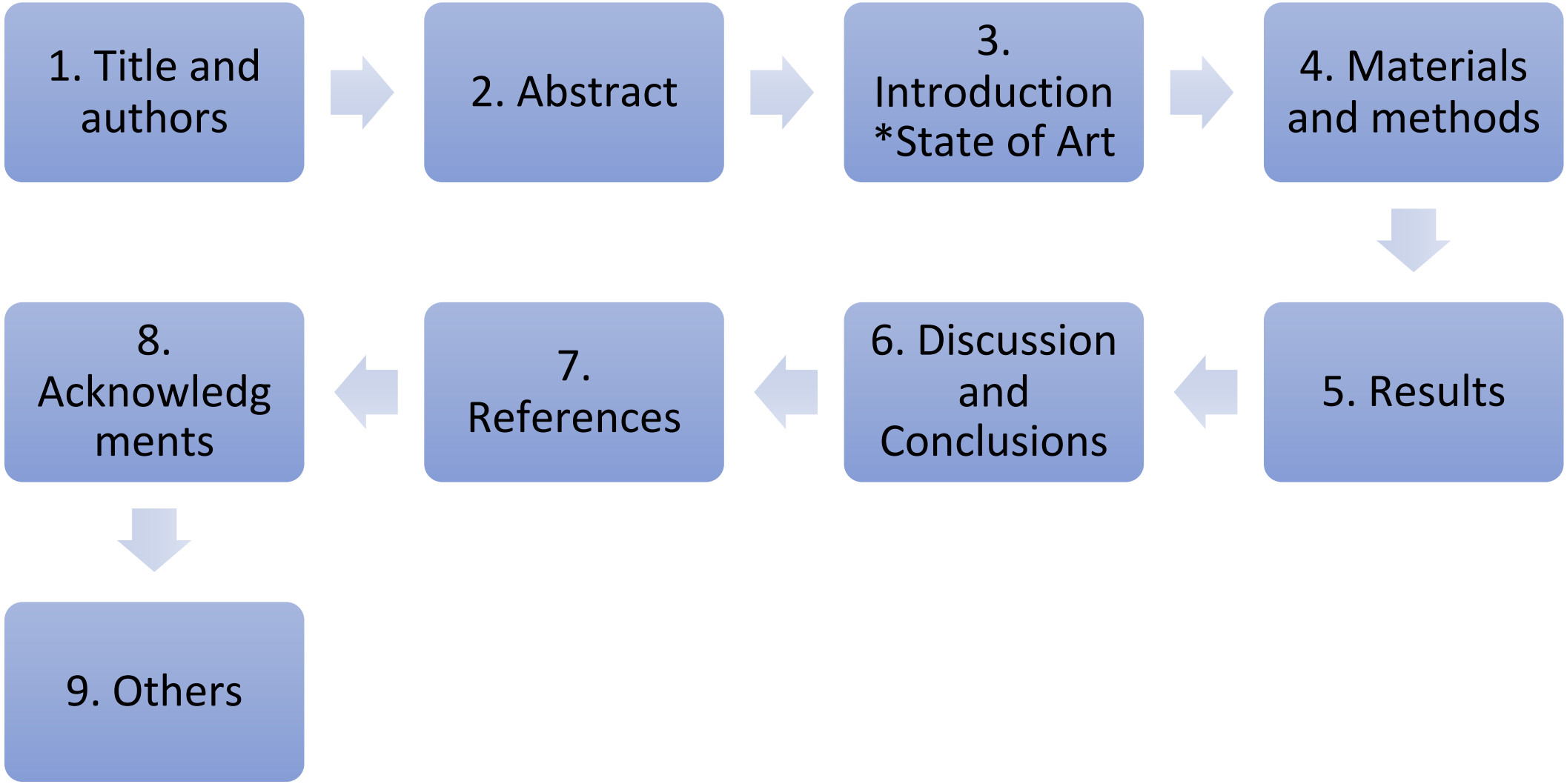
It should have a formal style.

It should be short and concise.

Be compatible with ethics.

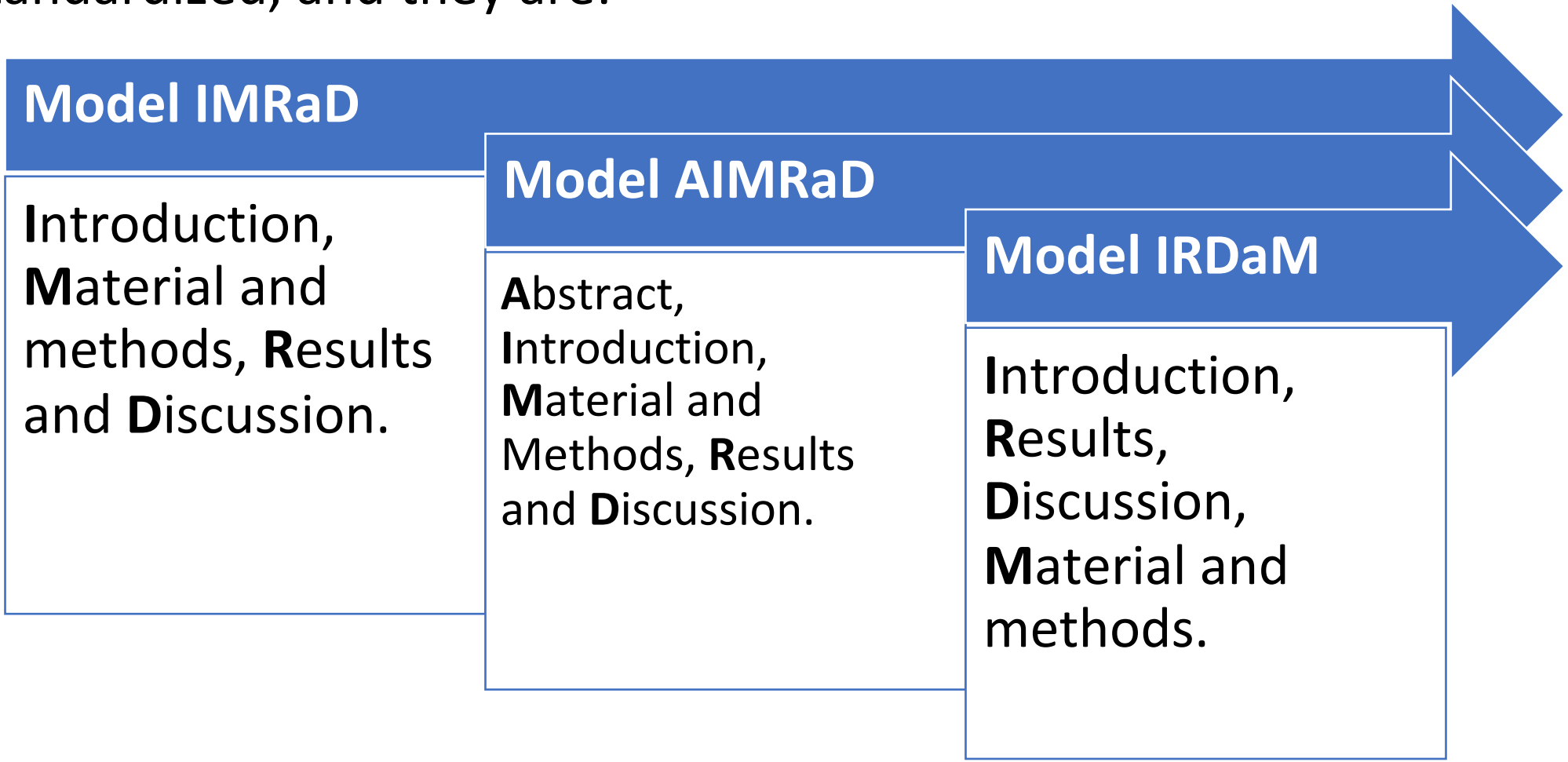
General structure of the scientific article

The scientific article is divided into sections, which are usually presented in the following order:



General structure of the scientific article

Nowadays there are different structures of organization of the articles that are standardized, and they are:



Title and authors

Characteristics

It should be short, concise, and clear.

It is recommended that it be written after writing the core of the paper (introduction, material and methods, results and discussion).

They can be informative or indicative.

Provides information on the scope of work. They can include part of the results.

Examples:

High incidence of myocardial infarctions in smokers (informative).

Incidence of myocardial infarction in smokers (indicative).

Below the title the authors, informative data and affiliation are described.

Title and authors

La integración social de los hijos de inmigrantes africanos, europeos del este y latinoamericanos: un estudio de caso en España*

Social Integration of Children of Immigrant Parents from Africa, Eastern Europe and Latin America: a Case Report in Spain

Recibido: 15 de mayo de 2012 | Revisado: 6 de octubre de 2014 | Aceptado: 11 de mayo de 2014

JUAN CARLOS CHECA**

Author Proof

Framing and Immigration: New Frames in Media and Social Networks

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risti
Revista Ibérica de Sistemas e Tecnologias de Informação
Iberian Journal of Information Systems and Technologies

Recebido/Submission: 03/04/2020
Aceitação/Acceptance: 16/06/2020

Agenda-setting e inmigración: análisis crítico del discurso y frecuencia en los medios: Estudio descriptivo de investigaciones en revistas científicas desde 2015 a 2020

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journalism and media

MDPI

Article

Analyzing Communication and Migration Perceptions Using Machine Learning: A Feature-Based Approach

Andrés Tirado-Espín [1], Ana Marcillo-Vera [1], Karen Cáceres-Benítez [2], Diego Almeida-Galárraga [2], Nathaly Orozco Garzón [3,*], Jefferson Alexander Moreno Guaicha [4] and Henry Carvajal Mora [3]

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[3] Faculty of Engineering and Applied Sciences, Networking and Telecommunications Engineering, ETEL Research Group, Universidad de Las Américas (UDLA), Quito 170503, Ecuador; henry.carvajal@udla.edu.ec
[4] Departamento de Posgrados, Campus Morelos, Universidad de Investigación e Innovación de México, Insurgentes 115, Centro, Jiutepec 62550, Morelos, Mexico; jmorenog@comunidad.uiix.edu.mx
* Correspondence: nathaly.orozco@udla.edu.ec

Abstract

Characteristics	
	It should allow the reader to identify the basic content of the work.
	Report all sections, including results and conclusions.
	Bibliographic references should not be included.
	The problem being investigated and its objective must be clear.
	It should state the main objectives and scope of the investigation.

Abstract

RESUMEN

La implementación de tecnologías de información como una herramienta para la toma de decisiones busca dar respuesta a la integración de un modelo Big Data en el Internet de las Cosas para la gestión de la producción de banano orgánico. Este trabajo establece como incide la integración de un modelo Big Data en el Internet de las Cosas (IoT) en la gestión de la producción de banano orgánico en la provincia de El Oro identificando un modelo que recolecta datos in situ de las variables asociadas a la producción de banano e integrando el Internet de las Cosas y big data. La investigación se desarrolló en base a un modelo Big Data Agrícola el cual analizó las variables relacionadas a los sensores, tecnología, procesamiento, analítica de los datos y toma de decisiones. El tipo de investigación es correlacional y descriptiva, aplicando técnicas documentales y de investigación de campo con un análisis estadístico para dar fundamento científico al estudio realizado. Los resultados establecieron cómo se integra la tecnología Big Data e Internet de las Cosas como apoyo a la gestión de la producción de banano orgánico, identificando las estrategias que facilitarán su implementación.

Palabras clave: Big Data, Internet de las Cosas, gestión agropecuaria, sensores, banano orgánico.

Vite Cevallos, H., Townsend Valencia, J., & Carvajal Romero, H. (2020). Big Data e Internet de las Cosas en la producción de banano orgánico. *Revista Universidad y Sociedad*, *12*(4), 192-200.

Introduction

Characteristics	
	Answer the question: **why this work has been done?**
	The interest of research in the scientific context is exposed.
	Previous work on the subject should be included and what aspects are unclear.
	The last paragraph of the introduction is used to summarize the objective of the study.

Hypothesis design



OBJETIVO GENERAL:

Analizar la influencia de los encuadres noticiosos en la percepción de la población juvenil española respeto a la inmigración. A partir de la contrastación entre métodos de investigación tradicional y técnicas neurocientíficas.

Objetivo Específico	Pregunta de Investigación (RQ1)
Analizar la relación que existe entre los encuadres noticiosos que abordan el tema de la migración y las emociones que experimentan los jóvenes españoles respecto al tema.	**RQ1: ¿Cuáles son las emociones que experimentan los sujetos nacionales al leer una noticia de migración con un encuadre negativo vs positivo?** Esta RQ se apoya en el estudio realizado por Cachón, D. (2016), en la cual se analiza los mecanismos explicativos del efecto *framing* y, concretamente, el papel mediador las emociones empáticas y el transporte narrativo en el efecto de las noticias. En este estudio se examinó la forma cómo encuadrar la información para favorecer reacciones empáticas y reducir emociones negativas hacia los inmigrantes en España.

Materials and Methods

Characteristics

This section answers the question: **how the study was done?**

It should be detailed enough so that another author can repeat the study and independently verify the results.

This section is organized as follows:

Materials and methods

1. Desing	• Randomized, controlled, cases, clinical trial, longitudinal, etc.
2. Population	• Sample and selection criteria.
3. Environment	• Habitat, place where the study takes place.
4. Interventions	• Techniques, measuring instruments and units, pilot tests, apparatus and technology, etc.
5. Statistic analysis	• It indicates the statistical methods used and how the data has been analyzed.

Results

Characteristics

They fulfill two functions:

1. Express the results of the experiments described in the Material and Methods.

2. Present the evidence that supports these results. Through figures, tables or in the same writing.

Discussion

Characteristics

This section is very important in the scientific article.

The results are interpreted, valued, commented and contextualized, in relation to the objectives, research questions and hypotheses.

The findings reached in relation to current knowledge on the subject of study are explained.

Discussion

Characteristics	The discussion begins with the answer to the question in the introduction with the evidence presented in the corroborating results.
	Bring out and clearly comment on the anomalous results, giving them as consistent an explanation as possible or simply saying that this is what you have found.
	Speculate and theorize with imagination and logic. This can pique your readers' interest.
	Avoid drawing more conclusions than your results allow, even if those conclusions are less spectacular than expected or desired.

Bibliographic references

Characteristics

They fulfill two functions:

1. Testify and authenticate the previous data of the work.
2. Provide relevant references for the topic of study.

It will be cited according to the regulations required by the chosen journal or the scientific Editorial.

The updating of the scientific article will be determined based on the bibliographies consulted, whose ideal validity is in **the last 5 years of publication.**

Acknowledgment

Characteristics

It is an optional section at the end of scientific articles.

Provides the opportunity for authors to mention individuals, groups, or institutions that helped with the study or preparation of the document.

They should not be given to people who are obviously involved in the execution of the research or the editorial process (ethics committees, research committees, editorial reviewers, the journal, the authors, etc.)

It must be differentiated from the financing (sponsors) and dedication.

Financing and dedication

Financing

Most magazines have a section for funding sources where you can put the name of the institution, the project code and even what was funded (total or partial project; scholarships for students; equipment, etc).

If this section is completed, it is redundant to rewrite the same in the acknowledgments.

In journals that do not have a clear policy and section for funding sources, the acknowledgments section is suitable for storing the funder's information.

Dedication

The dedication is an honorary declaration that the author makes to a person or entity of his work: in general, the person to whom the dedication is made did not participate in it.

This element is frequent in degree and thesis works; but, it is not used in scientific articles.

Ejemplo de agradecimiento

recognition between malignant and benign lesions, based on FE using ResNet50 and ResNet101, for a posterior feature selection using kurtosis controlled principle component (KcPCA) to feed an SVM classifier. They used HAM10000, ISBI 2017, and ISBI 2016 databases separately and reported 95.60% accuracy using the ISBI 2017 database.

In most scientific reports for skin lesion classification based on FE and SVM training, the computing time required to run each model is not included; therefore, it is difficult to compare this parameter. However, the methodology proposed in this study has achieved good accuracies in low periods.

The limitations presented in this study lie in the data amount used for training and validating the models. Here, 112

To sum up, it successfully developed ten different classification models, capable of detecting and classifying seven skin pathologies by using the dermatoscopic images from the HAM10000 database. The methodology proposed in this study resulted in efficient skin lesions classification methods with high accuracy and little time required for each of them.

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Other aspects

Characteristics

Conflict of interests.

In a certain manuscript when a participant (author, reviewer or editor) in the process of writing, reviewing and publishing; has links to activities that may inappropriately influence your judgment.

Conflict can also occur for other reasons, such as personal relationships, academic rivalry, or intellectual passion.

Public confidence in the peer review process and the credibility of published articles depend in part on how well conflict of interest is handled during manuscript writing, peer review, and decision-making by the editors.

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DEPARTAMENTO DE DESARROLLO HUMANO

Types of research

Andrés Tirado, Ph.D.

Types of research

Level of knowledge

- [ ] Exploratory
- [ ] Descriptive
- [ ] Correlational
- [ ] Explanatory

Degree of experimentation

- [ ] Experimental
- [ ] Quasi experimental

Measure characteristics

- [ ] Qualitative paradigm
- [ ] Quantitative paradigm

Exploratory Study

Characteristics:

- They are novel studies, little explored.
- Its results cannot be generalized.
- They are used to study a problem without a clear definition.
- Prioritize people's opinions:

- **What is the risk that is triggered by this action for the researcher?**



Methods:

- **Primary:** a problem is studied in depth.
  - Surveys
  - Interviews
  - Focus Groups.
  - Observation.
- **<u>Secondary: previously published research.</u>**
  - Case studies, online and bibliographic references.

Descriptive study



Characteristics:

- Its nature comes from its own name, *describe.*
- These studies emphasize: What? Unlike the others who seek the Why ?.
- Analyze sectors of the population.
- It is quantitative, it identifies various components of a phenomenon.

Methods:

- Observation: it does not influence any type of variable.
  - Qualitative: it does not involve numbers.
  - Quantitative: figures, data, quantities.
- **Case studies**: deep analysis of the study phenomena.
  - The data cannot be generalized. It does not determine cause and effect.
- **Surveys:** it is the most used technique with this type of investigation. Sample is great.

Correlational Study



Characteristics:

Causes and effects.
Why cause what?

- Positive effects => Enhance.
- Negative effects => Prevent.
- Use the correlations A => B
- Correlated facts are not necessarily related.
- Correlated facts may have a common cause.
- Either of the two correlated events can cause the other.
- The causes are complex.

Observation

The researcher does not influence the phenomenon of study. It takes place in a public environment

Surveys

Collect data.

Define the universe and the type of sample.

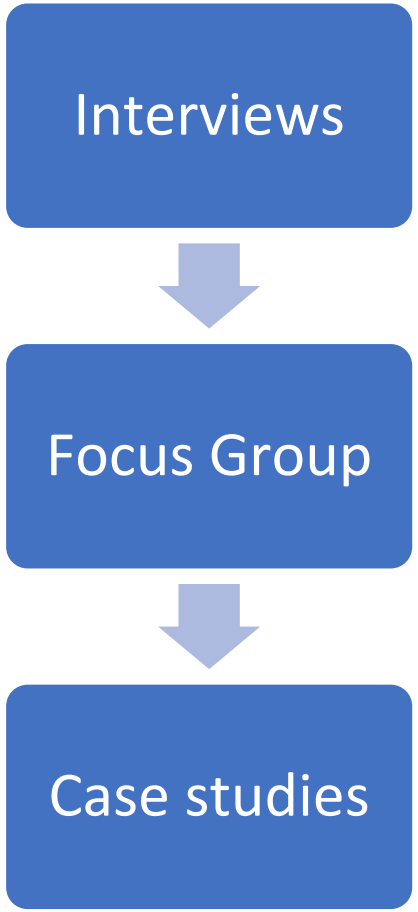
Data file

Previous studies are used.

Explanatory Study

Characteristics:

- They are used to explain a topic in detail.
- These studies are the preamble to a larger investigation.
- The data is not conclusive.
- From this type of studies, Research Questions are generated.
- They provide new perspectives to the phenomenon of study.

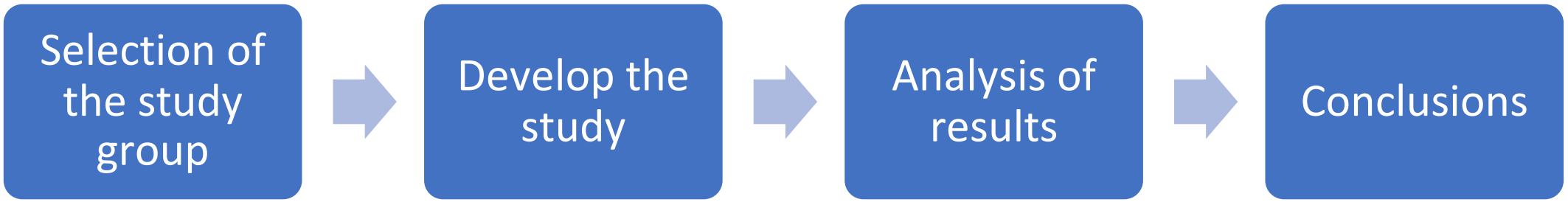


Quasi experimental

It is a type of study that is characterized in that the study subject is not selected randomly, but is found or established previously.

The methodology of this type of research is characterized by being descriptive.

Record qualitative and quantitative data.



Experimental

It is any investigation carried out with a scientific approach, where one set of variables is kept constant, while the other set of variables are measured as the subject of the experiment.



An experimental investigation is a laboratory test.

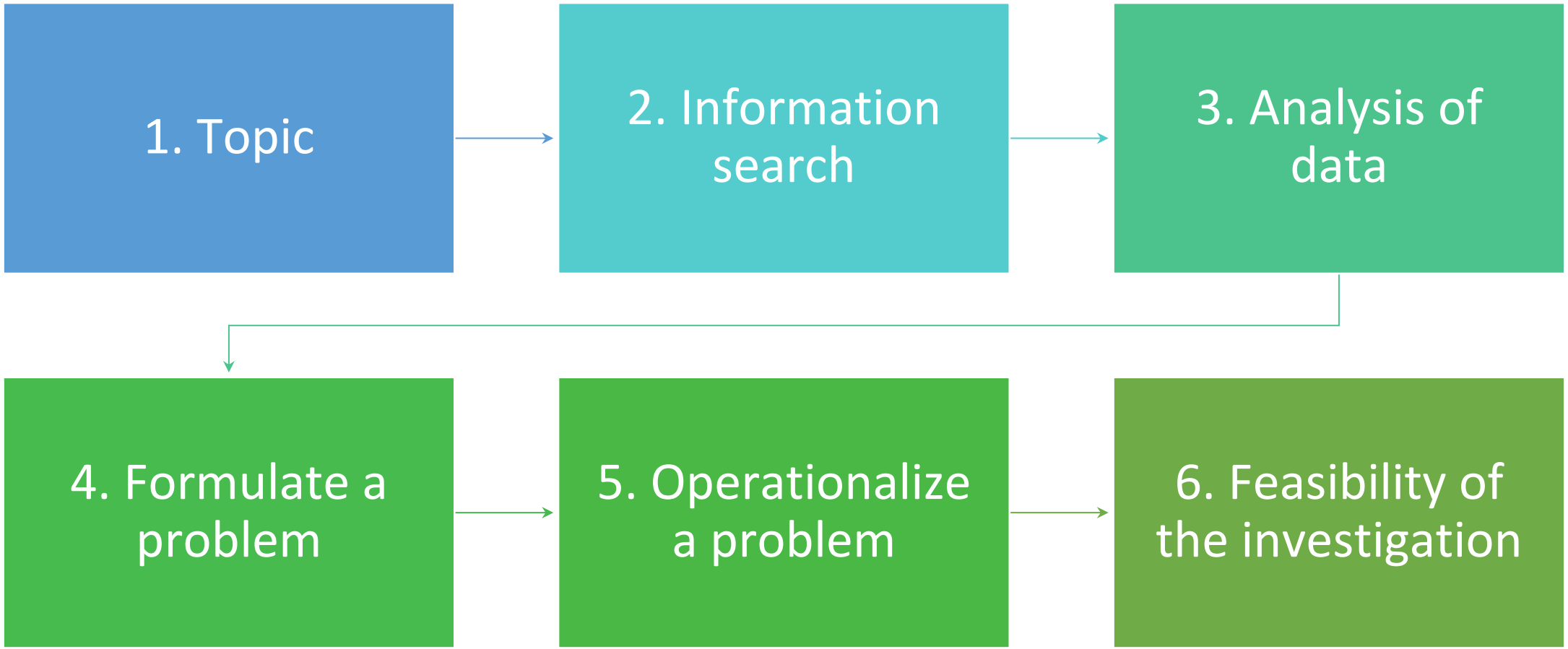
It is considered successful when there is a change in the dependent variable due to manipulation of the independent variable.

It is based on statistical analysis to prove or disprove a hypothesis.

Qualitative versus quantitative paradigm

Qualitative	Quantitative
Phenomenology	Positivism
Observation	Metrics
Subjective	Objetive
Investigator intervention	Distance with the study phenomenon
Emphasize process	Emphasize results
Valid: deep and real.	Safe: replicable.

Research Organization



Boolean Method



It is a method that is used to obtain more exact results on a specific topic.

1. Determinan descriptors: fixed and variable..
2. It is a binary system (0-1).
3. Formula is: 2^n = number of combinations depending on the descriptors. Ej. 2n = 2^5 = 32.

Boolean Method

Google Scholar Advance

Scopus

Web of Science

$2^n = 2^5 = 32$

Búsqueda Boolena								
Número	Agenda Setting	Inmigración	España	Opinión Pública	Jóvenes	Medios Digitales	Internet	Redes Sociales
1	1	1	1	0	0	0	0	0
2	1	1	1	0	0	0	0	1
3	1	1	1	0	0	0	1	0
4	1	1	1	0	0	0	1	1
5	1	1	1	0	0	1	0	0
6	1	1	1	0	0	1	0	1
7	1	1	1	0	1	1	1	0
8	1	1	1	0	1	1	1	1
9	1	1	1	0	1	0	0	0
10	1	1	1	0	1	0	0	1
11	1	1	1	0	1	0	1	0
12	1	1	1	0	1	0	1	1
13	1	1	1	0	1	1	0	0
14	1	1	1	0	1	1	0	1
15	1	1	1	0	0	1	1	0
16	1	1	1	0	0	1	1	1
17	1	1	1	1	0	0	0	0
18	1	1	1	1	0	0	0	1
19	1	1	1	1	0	0	1	0
20	1	1	1	1	0	0	1	1
21	1	1	1	1	0	1	0	0
22	1	1	1	1	0	1	0	1
23	1	1	1	1	1	1	1	0
24	1	1	1	1	1	1	1	1
25	1	1	1	1	1	0	0	0
26	1	1	1	1	1	0	0	1
27	1	1	1	1	1	0	1	0
28	1	1	1	1	1	0	1	1
29	1	1	1	1	1	1	0	0
30	1	1	1	1	1	1	0	1
31	1	1	1	1	1	1	1	0
32	1	1	1	1	1	1	1	1

Formulate a problem

1. Define a problem

- Discuss a specific topic.
- It is not only necessary to study problems in the negative sense of the term.
- To learn about problems or issues worth investigating, we must observe and read.
- You have to talk to experts and those involved.

2. Delimit the problem

- What we will study must be specified and clearly specified.
- The delimitation of the problem goes through specifying the action that takes place and the actors involved (they can be organisms, people, organizations, ecosystems, countries).
- It is recommended to specify the place and / or time of the problem.

Formulation Research Questions (RQ) and Hypotheses

It is the part in which the researcher generates an assumption that will later be confirmed or rejected once the research goes through experimentation and analysis of results.

Hypotheses

- Carry out a preliminary investigation on the topic covered.
- From the results of other investigations I am going to justify what I want to know.
- You need a theoretical support with bibliographic references.
- It is the answer to previous theories or studies.
- Measure the variables: dependent and independent.

Hypothesis formulation

"In a pandemic story, the negative news frame will cause a greater experience of negative emotions towards viewers and a lower intensity in positive emotions.“

VI: new frame

VD: emotions

"New parents with higher education are more relaxed in childbirth.“

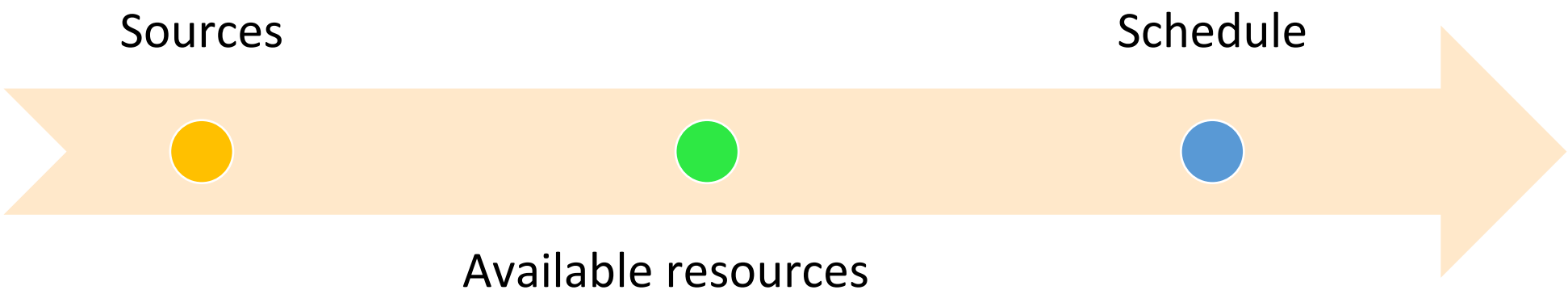
VI: Types of study

VD: attitude

Topic: Brand positioning of the mortiño wine "La Carmela", in Quito.
2.1. Ask yourself about one or more causes of the problem.
Why should they buy the product? What characteristics does the product have?
2.2. Wondering about the consequences of the problem in some area.
How does this product contribute to environmental care?
2.3. Think of a solution to the problem or issue. Wondering what would happen if we applied that solution to the bounded problem or how an action would affect the problem.
Will promoting the components of the wine generate sales and contribute to environmental care?

Topic: Brand positioning of the mortiño wine "La Carmela", in Quito.
2.4. Wondering if the problem happens elsewhere and wondering why or what consequences it has. In this way we formulate research questions for comparative research.
Why should consumers choose "La Carmela" wine over other types of organic wines?
2.5. Wondering if the current problem happened before, or if the past problem happens today. Thus we formulate questions for a longitudinal investigation.
Organic wines have become popular in recent years.

Research feasibility



Available resources

- Personal.
- Equipment for the development of research.
- Economy

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